1. Types of Data

Data is the collection of facts and information used for analysis, decision making, and research. In statistics, data is mainly divided into **Numerical Data** and **Categorical Data**.

1.1 Numerical Data

Numerical data represents quantitative values that can be measured or counted. These values are expressed in numbers and can be used to perform mathematical calculations such as averages, sums, and statistical analysis.

Numerical data can be further classified into:

Discrete Data – Data that consists of whole numbers and represents countable values.
Example: Family Size (number of family members in a household).

Continuous Data – Data that can take any value within a range and is usually measured.
Example: Household Income or Age of the household head.

Examples from the dataset:

- Age_of_Household_Head
- Household_Income
- Family_Size

These columns contain numeric values and are used for statistical calculations such as mean, variance, and standard deviation.

1.2 Categorical Data

Categorical data represents qualitative characteristics that describe categories or groups rather than numerical values. These variables classify data into distinct groups.

Examples from the dataset include:

- Education_Level (Primary, Secondary, Graduate, Post-Graduate)
- Owns_House (Yes / No)
- Urban_Rural (Urban / Rural)
- Household_ID (identifier)

Categorical variables are typically used for classification and grouping rather than mathematical calculations.

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2. Types of Statistics

Statistics is a branch of mathematics used to collect, organize, analyze, interpret, and present data. It is broadly divided into two categories:

1. Descriptive Statistics
2. Inferential Statistics

2.1 Descriptive Statistics

Descriptive statistics involves methods used to summarize and describe the main features of a dataset. It helps researchers understand the data by presenting it in a meaningful way through numbers, tables, and graphs.

Examples include:

- Mean
- Median
- Mode
- Range
- Variance
- Standard Deviation
- Graphs such as histograms and boxplots

In this project, descriptive statistics are used to analyze the distribution of household income, family size, and age of household heads.

2.2 Inferential Statistics

Inferential statistics involves using sample data to make predictions or generalizations about a larger population. It helps researchers draw conclusions beyond the data collected.

Examples include:

- Hypothesis testing
- Confidence intervals
- Regression analysis
- Predictive modeling

Unlike descriptive statistics, inferential statistics focuses on making predictions and decisions about populations based on sample data.

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3. Measures of Central Tendency

Measures of central tendency describe the central or typical value in a dataset. The three main measures are **Mean, Median, and Mode**.

3.1 Mean

The mean, commonly known as the average, is calculated by adding all values in a dataset and dividing by the total number of observations.

Formula:

Mean = (Sum of all values) / (Number of observations)

Example:
If five households have incomes of 30000, 40000, 35000, 50000, and 45000, the mean income would be:

Mean = (30000 + 40000 + 35000 + 50000 + 45000) / 5

The mean gives an overall average but can be affected by extreme values.

3.2 Median

The median is the middle value in an ordered dataset. It divides the dataset into two equal halves.

Steps to find median:

1. Arrange data in ascending order.
2. Identify the middle value.

If the number of observations is even, the median is calculated as the average of the two middle values.

Median is useful because it is not affected by extreme values or outliers.

3.3 Mode

The mode is the value that appears most frequently in a dataset.

Example:
If household income values are:

30000, 35000, 35000, 40000, 45000

The mode is **35000** because it appears most often.

Mode is particularly useful for categorical data.

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4. Measures of Dispersion

Measures of dispersion describe how spread out the data values are around the central value.

4.1 Range

Range is the difference between the highest and lowest values in a dataset.

Formula:

Range = Maximum Value − Minimum Value

Range provides a simple measure of data spread but does not consider all values in the dataset.

4.2 Variance

Variance measures how far each data point is from the mean and from every other data point in the dataset.

Formula:

$$\text{Variance} = \Sigma (x_i - \mu)^2 / N$$

Where:

- x_i represents individual data points
- μ represents the mean
- N represents the number of observations

A larger variance indicates greater variability in the data.

4.3 Standard Deviation

Standard deviation is the square root of variance and measures the average distance between each data point and the mean.

Formula:

Standard Deviation = √Variance

A small standard deviation indicates that values are close to the mean, while a large standard deviation indicates a wide spread of data.

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5. Gaussian Distribution (Normal Distribution)

A Gaussian distribution, also known as a normal distribution, is a symmetric probability distribution where most values cluster around the mean and fewer values appear as we move away from the center.

Characteristics:

- Bell-shaped curve
- Mean, median, and mode are equal
- Symmetrical around the mean

It is widely used in statistical analysis because many real-world variables follow this distribution.

Example:
Income distribution or test scores often follow a normal distribution pattern.

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6. Log-Normal Distribution

A log-normal distribution occurs when the logarithm of the variable follows a normal distribution.

Characteristics:

- Right-skewed distribution
- Values cannot be negative
- Common in economic data

Many financial and income-related datasets follow a log-normal distribution because income values cannot be negative and often have a few extremely large values.

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7. Three-Sigma Rule (Empirical Rule)

The three-sigma rule, also known as the empirical rule, describes the distribution of values in a normal distribution.

According to this rule:

- 68% of data lies within **1 standard deviation** from the mean
- 95% of data lies within **2 standard deviations**
- 99.7% of data lies within **3 standard deviations**

This rule helps identify outliers and understand data variability.

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8. Percentiles

Percentiles divide a dataset into 100 equal parts and indicate the relative standing of a value in a dataset.

For example:

- The 25th percentile means 25% of observations fall below that value.
- The 90th percentile means 90% of observations fall below that value.

Percentiles are commonly used in income analysis, exam scores, and performance evaluation.

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9. Quartiles

Quartiles divide the dataset into four equal parts.

The three main quartiles are:

Q1 (First Quartile) – 25% of the data falls below this value.
Q2 (Second Quartile) – The median of the dataset.
Q3 (Third Quartile) – 75% of the data falls below this value.

Quartiles help in understanding the spread and distribution of data.

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10. Five Number Summary

The five-number summary provides a quick overview of the distribution of a dataset.

It consists of:

1. Minimum Value
2. First Quartile (Q1)
3. Median (Q2)
4. Third Quartile (Q3)
5. Maximum Value

This summary is often used to construct **boxplots**.

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11. Skewness

Skewness measures the asymmetry of a distribution.

Types of skewness:

Positive Skew (Right Skew)
The tail extends toward the right side of the distribution.

Negative Skew (Left Skew)
The tail extends toward the left side.

Zero Skew
The distribution is perfectly symmetrical.

In income datasets, positive skewness is common because a few households may have significantly higher income.

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12. Kurtosis

Kurtosis measures the "tailedness" or shape of the distribution.

Types include:

Leptokurtic
High peak with heavy tails.

Mesokurtic
Normal distribution shape.

Platykurtic
Flatter distribution with lighter tails.

Kurtosis helps identify whether the dataset has extreme values or outliers.